



How to present collision warnings at intersections?—A comparison of different approaches

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ABSTRACT

Analyses of intersection accidents have shown that in most cases drivers overlook or see other road users with the right of way too late that they cannot react in time. Appropriate visual warning signals which support drivers in their attention allocation and driving behavior could be useful to improve drivers' reaction. In a driving simulator study, two warning strategies varying in their timing: (1) top-down warning while approaching a critical intersection and (2) bottom-up warning directly before the critical incident were investigated in a T-intersection situation. For the bottom-up warning, two warning signal designs were compared. 48 subjects ($M = 27.3$ years, $SD = 7.4$ years) participated in the study. Driving data as well as subjective evaluation of the three warning signals (one early and two late warning signals) were analyzed. The early warning signal which was given while approaching the intersection showed a positive effect. Here, most collisions could be avoided due to drivers' adaptation of their driving behavior toward safer driving. They waited longer at the intersection before turning and finally turned with a lower velocity. In addition, drivers evaluated the early warning signal as very "useful". With regard to the late warnings a much smaller effect was found. From these results, requirements can be derived for the design of effective warning strategies when driving at critical intersections.

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1. Introduction

Driving at intersections is one of the most dynamic and complex task of drivers. Depending on the intersection situation (e.g. intersection type, driving maneuver, and traffic volume), drivers have to perceive different kinds of information by shifting their attention to various task-relevant objects, e.g. crossing vehicles, vehicles driving ahead and oncoming, traffic signs, as well as pedestrians and cyclists. For safe driving, a high degree of visual and spatial processing is required (Hills, 1980; see also, Cummings, 1986; Moore, 1969). In-depth analyses of car accidents have found that perceptual errors which are also called recognition errors (Rumar, 1990) are one of the major contributing factors for accidents at intersections (see, e.g. Clayton, 1976; Fell, 1976; Sabey and Staughton, 1975; Staughton and Storie, 1977; Treat et al., 1979; Vollrath et al., 2006). For example, an in-depth accident survey in Germany (Vollrath et al., 2006) found that in above 90% of intersection accidents lack of information was the main reason leading to an accident. In most cases, drivers failed to yield to other traffic participants with the right of way as they were not noticed by the

drivers. Other accident studies have shown similar results (see, e.g. Graab et al., 2008; Gründl, 2005; Hoppe et al., 2007; Wierwille and Tijerina, 1995).

In most of the intersection accidents mentioned above, drivers overlooked other traffic participants ("Looked-but-failed-to-see accidents"; see Brown, 2005; Cairney and Catchpole, 1996; Herslund and Jørgensen, 2003; Koustanai et al., 2008; Langham et al., 2002; Treat, 1980; White and Caird, 2010) or failed to look at the right place and thus saw the other traffic participant too late to react in time. Especially, according to different in-depth studies in the Nordic countries (see, e.g. Pasanen, 1992; Räsänen and Summala, 1998, 2000; Summala et al., 1996) this last kind of perceptual error was found to be relevant in bicycle-car accidents. According to the authors, improper visual scanning strategies and unjustified expectations for events and hazards by drivers could be responsible for this.

With regard to the second accident type ("failed to look"), a recent driving simulator study by Werneke and Vollrath (2012) confirmed these results. The aim of the study was to investigate drivers' attention allocation and driving behavior in different T-intersection situations which varied in traffic volume and the presence or absence of pedestrians. Additionally, it was examined how well drivers could react to a critical event presented in these different intersection situations. Results of the study showed that, first with regard to the intersection situation drivers allocated their

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attention mainly to certain areas of the intersection where they expected task-relevant objects. In contrast, other areas were rather neglected by them.

Second, their velocity when turning differed depending on the intersection situation. Especially, at intersections with a low traffic volume and no pedestrians they drove with the highest velocity. This resulted in many collisions when a critical event occurred. In contrast, drivers could best avoid a collision when they allocated their attention more properly at another type of intersection and turned right with a lower velocity.

In conclusion, there are certain intersection situations in which the probability of an accident is systematically higher because of drivers' improper attention allocation and driving behavior. As this is the case, one can try to prevent these accidents by using appropriate driver warning systems which support drivers in their attention allocation and driving behavior.

In the last decade, a lot of effort was put into the development of intersection assistance systems (see, e.g. Doerzaph et al., 2007; Hopstock et al., 2005; Mages, 2008; to mention just a few) and into the design of appropriate human-machine-interface concepts (see, e.g. Benmimoun et al., 2007; Campbell et al., 2007; Klanner et al., 2008; Thoma et al., 2008). However, there have only been few studies which systematically compared different strategies which aim at supporting drivers in the most appropriate manner (e.g. Lee et al., 2009). One warning strategy could be to try to inform drivers at an early stage about a potential critical event at the intersection so that drivers can adapt their attention and driving behavior well in advance in a manner that enables them to better react when a critical situation occurs. The second warning strategy could be to warn drivers immediately before the critical event so that they can react faster when seeing the critical object.

The aim of the first warning strategy can be described as trying to change drivers' expectations about the current driving situation so that they are prepared for the situation and can adapt their attention and driving behavior accordingly. From a technical point of view this requires knowledge about future events or about traffic participants which are not directly visible for the driver. Car-to-x communication technologies which are currently being developed could provide this kind of information (see, e.g. Festag et al., 2008; Franz, 2004; Lübke, 2004; Röglinger, 2011; Röglinger and Facchi, 2009).

With regard to the second warning strategy, one would try to direct drivers' attention toward the critical event in order to elicit a fast and effective reaction. One could present warning signals in the focus of drivers' view (e.g. in a head-up display) so that the warning is easily perceived and drivers are enabled to react quite fast. One could also try to present salient and powerful warning signals in drivers' peripheral vision which attract their attention in an automatic manner (see, e.g. Egeth and Yantis, 1997; Remington et al., 1992; Ruz and Lupiáñez, 2002; Theeuwes, 1992, 2004; Theeuwes

and Godijn, 2001; Yantis and Egeth, 1999; Yantis and Jonides, 1990). This would disrupt the current attention allocation strategy of drivers and would direct their attention toward the appropriate intersection area so the unexpected hazard can be perceived faster. As a result, drivers can react faster. Additionally, this does not require conscious awareness by drivers and does not reduce their capacity for performing other tasks (see, e.g. Jonides, 1981; Posner and Snyder, 1975).

First studies have already demonstrated the positive effect of such peripheral warning signals (see, e.g. Arnon, 2010; Falkmer et al., 2000; Fricke, 2009; Henning et al., 2008; Ho and Spence, 2009; Kienast et al., 2008; Laquai et al., 2010; Mahlke et al., 2007; Victor and Jarlengrip, 2005). In most cases, peripheral warning signals were examined in context of pedestrian detection by drivers at night (see, e.g. Henning et al., 2008; Kienast et al., 2008; Mahlke et al., 2007). While these peripheral in-vehicle warning signals have been shown to be effective, it is still unclear how these signals should be best designed and what are the most appropriate signals especially in complex intersection situations where there is already a lot of information presented for drivers.

To summarize, theoretical considerations suggest that two different warning strategies might be effective to prevent intersection accidents where drivers overlook other traffic participants or see them too late. On the one hand, an early warning could enable drivers to adapt their attention and behavior in a manner that critical objects are better detected and drivers are able to react faster. On the other hand, later warnings presented concurrently with the appearance of critical objects could support drivers to perceive these objects and react faster. These late warnings could either be presented in the drivers' central view which ensures that the warning can be perceived very well by them. However, drivers would then still have the problem of perceiving the critical object. In order to facilitate this, one could also present peripheral warnings which are able to attract drivers' attention toward the critical object.

As expectations allow drivers to adapt their behaviors, one would expect the warning strategy (first warning strategy) to be most effective. However, it is not always feasible to have this kind of information well in advance. Thus, the second warning strategy (late warning strategy) should also be examined. As it takes time to shift the attention one would expect that peripheral warnings should be more effective as warnings presented in the center of drivers' view. These peripheral warnings lead to an automatic and very fast shift of attention. In addition, it should take some additional time to understand a warning which is presented in the middle of the field of view.

In line with these considerations, three warning signals were compared in the present study. To this aim, an unsignalized T-intersection situation was used in which drivers had to turn right and where a critical event happened just after drivers had begun their turn. The three warning signals used in the study are described in detail in the next section.

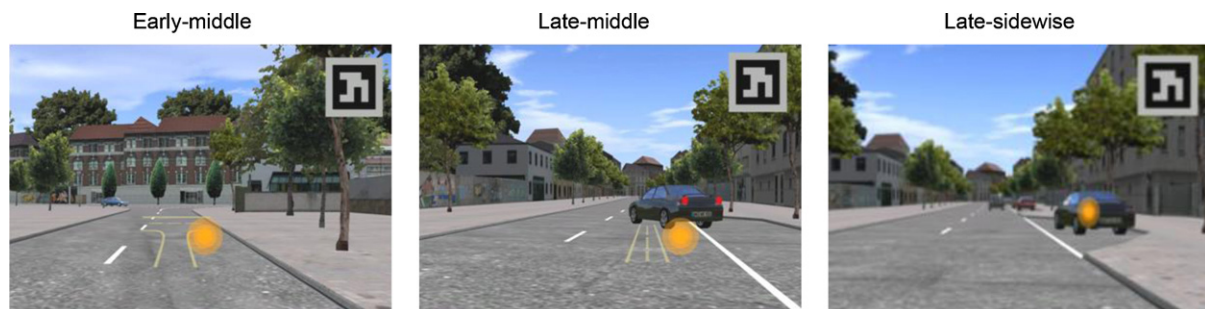


Fig. 1. The three warning signals examined in the driving simulator study: Top-down, early-middle warning signal presented in HUD (picture left); bottom-up, late-middle warning signal presented in HUD (picture middle); bottom-up, late-sidewise warning signal presented in augmented reality (picture right).

2. Method

2.1. In-vehicle warning signals

In the study, three warning signals were investigated which are shown in Fig. 1. All signals were presented in a simulated head-up display (HUD). The critical incident at the intersection was always a car parked at the right-hand side of the road which suddenly left the parking space and entered the road just as drivers were actively executing their right turn (see Fig. 2). All warning signals were continuously presented to drivers by the end of the critical incident.

As Fig. 1 shows, the early-middle warning signal was presented to drivers while approaching the critical intersection situation (approximately 70 m before the limit line of the intersection). It used a bird's eye view of the intersection where a flashing orange warning circle signalized drivers the intersection side where a potential critical incident would occur.

The aim of the early warning was to change drivers' expectations while approaching the critical intersection situation (top-down processes). With this early warning signal drivers could expect that certain areas of the intersection could be critical so that they could allocate their attention to these intersection areas. Thus, drivers were prepared for danger and could attend their attention appropriate to the current driving situation. In order to achieve that goal, the warning was given in advance at an early stage.

The two late warnings (Fig. 1) occurred approximately at the same time when the parked car suddenly started to enter the road (approximately 18.5 m away from the entering car when drivers passed the limit line). In the late-middle warning signal also a bird's eye view of the road was shown to drivers. A flashing orange warning circle symbolized the road side at which the critical incident was presented (here on the right-hand side of the road). The late-sidewise warning signal did not show the road. This signal was presented in augmented reality to drivers at the spot where the critical incident occurred. As both signals appeared quite suddenly in signal colors, they were supposed to capture drivers' attention.

The aim of the late warnings was to reallocate drivers' attention toward the intersection area with the critical incident by using

attentional bottom-up processes, e.g. by presenting salient and powerful warning signals in drivers' peripheral vision. The goal was to disrupt the current attention allocation by drivers and direct their attention toward the critical incident. With regard to the late-middle warning (Fig. 1), the signal design could make it easier to be noticed and seen by drivers as it is more directly within drivers' view. The aim of the late-sidewise warning was to shift drivers' attention directly to the critical event (peripheral warning). However, the bottom-up warning strategy is only useful when the warnings are given near the critical event at a point in time when the critical incident can be perceived by drivers.

2.2. Participants

A total of 48 subjects (16 female, 32 male) participated in the study. Participants ranged from 21 to 56 years of age ($M = 27.3$ years, $SD = 7.4$ years) and had a driver's license for 9.0 years ($SD = 6.6$ years). With regard to their behavior of journeys they drove about 45% in urban areas ($SD = 21.6\%$), 26% on rural roads ($SD = 17.4\%$), and 29% on freeways ($SD = 19.7\%$). There were four experimental groups (control group without a warning signal, group with the early-middle, late-middle, and late-sidewise warning signal). An equal number of female ($n = 4$) and male ($n = 8$) drivers were assigned to each experimental group.

Participants were recruited from the TU Braunschweig and in local supermarkets. All were extensively trained in a fixed base driving simulator. This training was conducted on a separate day with duration of approximately 1.5 h to avoid simulator sickness. This was necessary as driving in simulated urban scenarios with turning at intersections is hard for untrained participants. The training contained three driving scenarios. In the first driving scenario, the participants drove straight on a rural road to become familiar with the dynamics of the driving simulator. In the second scenario, acceleration and deceleration were trained with different velocities. Finally, the participants had to turn right and left at different intersection types. They had to drive through a large number of turns at low speed (50 km/h) in an urban environment. This can be in particular unpleasant for participants as visual and kinesthetic cues differ quite largely from real driving. Participants who developed simulator sickness had to be excluded from the study ($n = 25$ out of overall 73 trained participants). In the study, all remaining participants ($n = 48$) had normal or corrected-to-normal vision. They were compensated for their time either by receiving credits (students at the TU Braunschweig) or 30 Euros for the successful participation.

2.3. Driving scenario

For examining the effectiveness of the three warning signals (Fig. 1) in order to reduce or even avoid a collision in a critical intersection situation, a situation with a high collision frequency was required. This had been identified in a driving simulator study by Werneke and Vollrath (2012) mentioned above. Thus, a T-intersection situation with a yield sign was chosen where drivers had to turn right and had to yield to oncoming cars from the left-hand side of the intersection and driving straight (Fig. 2). The traffic density of the oncoming cars from left was low which indicated approximately nine cars in 1 km. Except for the oncoming cars from the left there were no other special features (e.g. pedestrians) in the intersection situation. Thus, the situation indicated to be an easy situation for drivers. As a result, they did not have to stop at the intersection before turning right. Overall, this specific intersection situation can be used to examine the effectiveness of the three warning signals (Fig. 1) for avoiding or at least reducing the consequences of a collision.

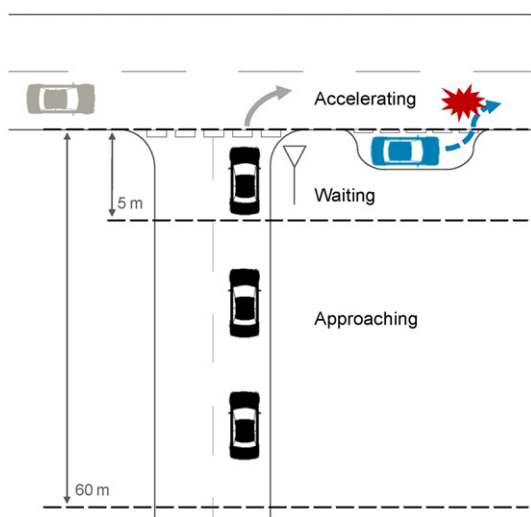


Fig. 2. The unsignalized T-intersection situation with the host vehicle who has to turn right (black car), with a crossing vehicle coming from the left-hand side and driving straight (gray car), and the vehicle as critical incident parking on the right-hand side of the road. The latter vehicle suddenly started and entered the road when drivers had passed the limit line, and thus began to enter the intersection. Additionally, the three analysis epochs of the driving data are illustrated: Approaching, Waiting, and Accelerating.

With regard to the critical incident, an unexpected hazard with a high collision frequency from the previous study by Werneke and Vollrath (2012) was chosen. A vehicle parked at the right-hand side of the road suddenly left the parking space and entered the road whilst drivers were actively executing the right turn (see Fig. 2). The entering vehicle was approximately 18.5 m away from the drivers when they entered the limit line of the intersection. This critical incident was presented two times to drivers in one driving scenario. In Fig. 2, the unsignalized T-intersection situation with the entering vehicle as critical incident is illustrated for the description of the driving data analysis used in the study.

Every participant drove one driving scenario. The scenario consisted of a number of urban segments. These segments composed of different T-intersection situations similar to the critical intersection situation. These intersection situations varied in the traffic density of oncoming cars from the left (low and high traffic density) and other road users (with or without pedestrians on the right-hand side). This was done to familiar drivers with T-intersection situations. In these uncritical intersection situations, there was also a parking vehicle on the right-hand side of the intersection, but which not entered the road. Each urban segment consisting of the T-intersections was connected with a short rural road.

The driving scenario had a length of approximately 50 km (about 31 miles) and took 45 till 50 min. At the beginning of the driving scenario, drivers passed through the urban segments in which no critical event occurred. After approximately 25 min, in one urban segment the critical intersection situation (Fig. 2) with the suddenly entering vehicle occurred. Afterwards, drivers continued the driving and repeatedly passed through further urban segments consisting of the uncritical T-intersection situations. At the end of the driving scenario, the suddenly entering vehicle (Fig. 2) occurred a second time. This was done to examine if drivers will adapt their driving and gaze behavior over time. However, results of the collision frequencies showed that in the second critical event, except of one participant all ($n=47$) could avoid a collision. As a result, further analyses were only conducted on the first time when the critical event occurred. Only this situation is used to examine the effectiveness of the three warning signals compared to the control group.

2.4. Experimental design and dependent variables

The experiment was a between-subjects design with four experimental groups as between-subjects factor: (1) control group without a warning signal, (2) group with the early-middle, (3) with the late-middle, and (4) with the late-sidewise warning signal. The participants were randomly assigned to one of the four experimental groups.

With regard to the dependent variables, driving and gaze data in the two critical incident situations were recorded. Additionally, subjective evaluations of the warning signal (Fig. 1) which was driven in the group were collected after the driving scenario. Because of the complexity of the study and data, this paper focuses only on the effectiveness of the three warning signals in the context of collision avoidance and drivers' driving behavior in the first critical situation as well as on drivers' subjective evaluation of the different warnings.

With regard to the collision avoidance in the critical situation the collision frequency (yes/no) and the minimal distance (in meter) to the entering vehicle were analyzed. Additionally, drivers' time-to-collision (TTC) and their brake reaction time (in seconds) were computed. The drivers' TTC in the four experimental groups was measured from the point in time of the late warning signals activation until the critical vehicle entered the road. This explains the time drivers would need when they drive on with the current

velocity, measured at this activation point. Drivers' brake reaction time was the time drivers started to brake after the late warning signals were presented.

To better understand drivers' reactions in the critical situation, their driving behavior was evaluated statistically in three epochs (Fig. 2): (1) *Approaching*, in which drivers moved straight forward while approaching the intersection, (2) *Waiting*, while drivers waited at the limit line and prepared to make a right turn, and (3) *Accelerating* point, in which in which drivers entered the intersection and executed the right turn.

The *Approaching* epoch was defined beginning 60 m before the intersection until drivers stopped or nearly stopped (velocity < 5 km/h) at the intersection. This was done to determine drivers' reaction to the early-middle warning signal (Fig. 1, left) which was presented to drivers approximately 70 m before the limit line of the intersection. The second epoch, the *Waiting* epoch, was defined beginning with the end of the *Approaching* epoch and ending when drivers began to enter the intersection (velocity ≥ 5 km/h). As the critical intersection situation indicated to be an easy situation for drivers, not every driver stopped before the limit line. In most cases, drivers only decelerated before turning right without really stopping. This was considered in the data analyses. The *Accelerating* point was approximately 7 m after the limit line. This limit was set to capture the initial phase of entering the intersection before the critical event occurred.

During the *Approaching* epoch drivers' velocity (in km/h) was analyzed at five distances before the limit line (60 m, 40 m, 20 m, 10 m, and 5 m) in order to estimate the speed course when driving with the early-middle warning signal, as compared to driving without a warning signal in this epoch (no signal in the control group, group with late-middle, and late-sidewise warning signal). While *Waiting* at the intersection, the waiting time (duration of waiting in seconds) was analyzed. As an indicator of how drivers entered the intersection, the velocity (in km/h) at the *Accelerating* point mentioned above was computed. The *Waiting* and *Accelerating* data were also analyzed by comparing the group with the early-middle warning with the group driving without a warning signal in these epochs (group with no signal, group with late-middle, and late-sidewise warning signal). Overall, all vehicle reactions were recorded by simulation software SILAB (Krüger et al., 2005) and were collected at a rate of 100 Hz.

The subjective evaluation of the warning signal was done in each group at the end of the driving scenario. The questionnaire assessed benefit, features, and design of the different warning signals. Besides a few open questions, mainly closed questions were used. For the rating, a 15-point rating scale was used (Heller, 1982). The rating scale consisted of five main categories in which each main category comprised 3-point scale ($-/0/+$) to refine the assessment. For example, when assessing the design of the warning signal, in a first step a scale of 1 (very bad) to 5 (very well) was used. In a second step, the participants used the 3-point scale ($-/0/+$) to refine their rating within this broader category.

2.5. Procedure

For the study, the fixed base driving simulator at the TU Braunschweig was used. At the beginning of the three hours session, the drivers were instructed in written form about the examination. All drivers were briefed to wear their own glasses if necessary in order to ensure normal vision. After reading and signing a consent form, two short driving scenarios from the training session were driven to get familiar with the driving simulator.

Next, the instruction of the driving scenario was given and one of the three warning signals (not in the control group) was explained to the drivers as an integrated warning system of the driving simulator. The drivers were instructed in written form about the

meaning and aim of the respective warning signal. If the warning signal should occur during driving, they should not be irritated by the warning signal. In a demonstration session, the warning signal was presented to drivers in a critical incident scenario (a pedestrian walking on the sidewalk and suddenly crosses the road). In order to deflect from the warning system and to avoid drivers' expectation of critical events in the driving scenario, they were told that we were interested in driving and gaze behavior in different intersection situations. As a result, the eye-tracking system was described. There was no mention of critical events to avoid participants waiting for such an event. In a next step, the eye-tracking system was put on and the calibration was carried out. The drivers were instructed to drive and to respond to traffic as they normally would in real environment. They had to observe and to obey traffic rules. The driving scenario lasted between 45 and 50 min.

During the whole experiment the investigator seated in a separate room. The investigator checked the driving simulation and eye-tracking system. If small recalibrations of the eye-tracking system were needed, they were made during driving. After the driving scenario, a short break was made and the actual aim of the study was explained to drivers. Then, they filled in the extensive evaluation questionnaire of the appropriate warning signal (not in the control group). After that, the other two warning signals (or all three warning signals in the control group) were explained to drivers and driven by them again only in the critical incident situation (Fig. 2). This enabled them to compare all three warning signals by using a short version of the evaluation questionnaire. At the end of the examination, the drivers were compensated and thanked for their participation.

3. Results

3.1. Data analysis

Results of the collision frequency in the first and second critical event showed that in the second time, except of one driver all out of the drivers could avoid a collision. As a result, further analyses were only conducted on the first time when the critical event occurred. Only this situation is used to examine drivers' driving behavior and the effectiveness of the three warning signals (Fig. 1) compared to the control group. In this section, the results of the first critical event are presented. For all data analyses the statistical software PASW Statistics 18 was used.

In order to examine drivers' reactions to the first critical event, collision frequency and drivers' minimal distance to the entering vehicle were analyzed. With regard to the collision frequency (yes/no), a Pearson's chi-squared test was conducted in order to compare the collision frequency between the four experimental groups. For the analysis of drivers' minimal distance (in meter) to the entering vehicle, an one-way analysis of variance (ANOVA) was conducted, using the groups as between-subjects factor. Additionally, an ANOVA was computed for TTC (in seconds) and drivers' brake reaction time (in seconds) in the first critical event.

Further analyses of drivers' driving behavior were conducted with regard to the three epochs mentioned above: *Approaching*, *Waiting*, and *Accelerating* (Fig. 2). In order to examine the drivers' speed courses while approaching the critical intersection, an one-way ANOVA for each distance to the limit line of the intersection (60 m, 40 m, 20 m, 10 m, and 5 m) was conducted. As no warning signals were issued at this point in time for the control group and the two groups with the late warnings, in the analysis all three conditions were considered as one group. The same analysis was done for the waiting time (*Waiting* epoch) as well as for the drivers' velocity while accelerating at the intersection (*Accelerating* point). In these later two epochs, no warning signals were issued for the

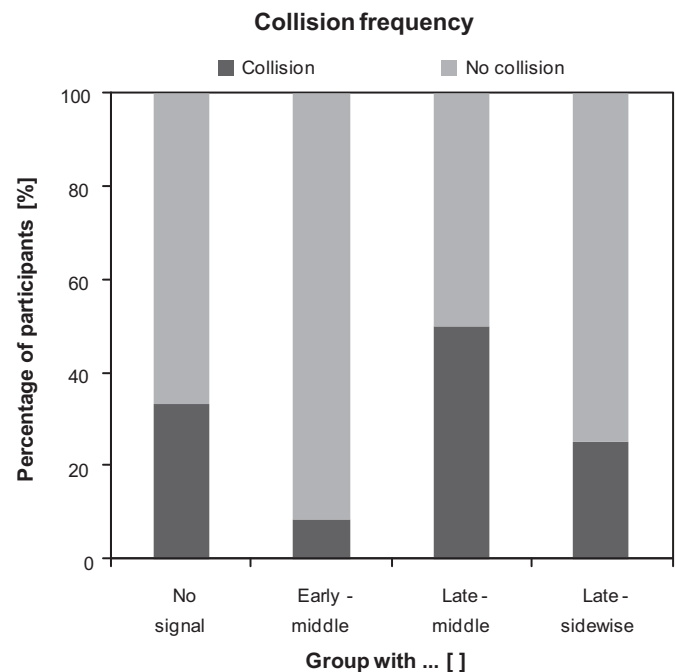


Fig. 3. Collision frequency (yes/no) at the first critical event, subdivided into the four experimental groups. Each bar corresponds to a sample size of 12 participants.

three groups. Additionally, previous analyses have shown that the three groups (control group and the two groups with the late warnings) did not differentiate from each other in the three epochs (*Approaching*, *Waiting*, and *Accelerating*), so they were combined as one group in the analyses.

For the subjective evaluation of the three warning signals with regard to their benefit, features, and design, for each evaluation criteria an one-way ANOVA was computed, using the warning signals as between-subjects factor.

In each analysis an alpha level of .05 was used. Results were corrected for sphericity violations where necessary by use of the Greenhouse–Geisser modification (Greenhouse and Geisser, 1959). In the following, the results of the drivers' driving behavior in the first critical event and their evaluation of the three warning signals are shown.

3.2. Results

3.2.1. Driving behavior

The analysis of the collision frequency in the first critical event showed no significant differences between the four groups ($\chi^2_{(3)} = 5.2$, $p = .155$). Looking at the pairwise comparisons, there were significantly fewer collisions in the group with the early-middle warning signal, as compared to the group with the late-middle warning ($p = .025$). The difference of the group with the early-middle warning signal to the control group was not significant ($p = .132$). Fig. 3 shows the frequency of collisions in the first critical event for the four experimental groups.

Fig. 3 shows that the fewest collisions occurred in the group with the early-middle warning signal. Except of one driver all could avoid a collision ($n = 11$ out of 12). Results of the group with the late-sideways warning signal showed that 75% of the drivers could avoid a collision ($n = 9$ out of 12). However, in the group in which the drivers were supported by the late-middle warning signal the collision frequency was quite high. Only half of the drivers ($n = 6$ out of 12) could avoid a collision. In contrast to this, in the control group without any warning signal, 75% of the drivers could avoid a collision ($n = 8$ out of 12).

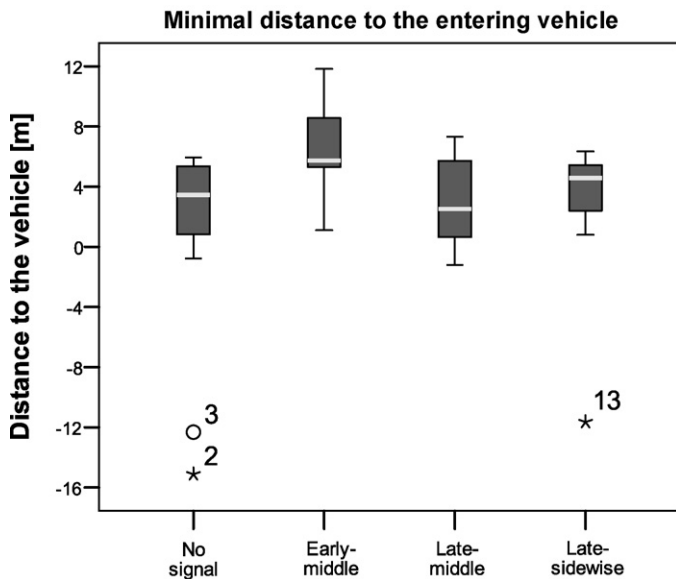


Fig. 4. Box plot of the minimal distance to the entering vehicle (in meter) in the first critical event, subdivided into the four experimental groups.

With regard to the drivers' minimal distance to the entering vehicle, a significant main effect of the group factor was found ($F_{(3, 44)} = 2.9, p = .046$). In Fig. 4 the drivers' minimal distance to the entering vehicle in the first critical event is shown for the four experimental groups.

As Fig. 4 shows, in the group with the early-middle warning signal the largest minimal distance was found (pairwise comparison to the control group: $p = .006$; pairwise comparison to the group with the late-middle warning signal: $p = .082$; pairwise comparison to the group with the late-sidewise warning signal: $p = .073$). The other pairwise comparisons were not significant. On average drivers with the early-middle warning signal stopped 6.5 m ($SD = 2.8$ m) before the entering vehicle. Drivers with the later warnings showed similar pattern of the minimal distance to the entering vehicle. Both groups had a mean minimal distance of approximately 3.0 m. However, differences in the standard deviation of the minimal distances were found. Whereas, in the group with the late-middle warning signal the standard deviation was approximately 3.0 m; in the group with the late-sidewise warning signal it was approximately 4.9 m. In the control group the lowest minimal distance was found ($M = 0.9$ m, $SD = 7.1$ m). Interestingly here, the standard deviation was very high compared to the other three groups which indicates that the drivers reacted differently to the critical event.

The analysis of the drivers' TTC (in seconds) to the entering vehicle showed a trend of a main effect between the groups ($F_{(3, 44)} = 2.2, p = .099$). The TTC of the group with the early-middle warning signal was by trend larger compared to the other three groups (pairwise comparison to the control group: $p = .091$; to the group with the late-middle warning signal: $p = .017$; to the group with the late-sidewise warning signal: $p = .093$). Drivers warned by the early-middle warning signal had a mean TTC of over 4 s ($SD = 1.5$ s). In the other three groups, the TTC was between 3.2 s and 3.5 s. As a result of a longer TTC in the group with the early-middle warning signal, drivers took significant more time to brake for the critical event (group: $F_{(3, 44)} = 4.7, p = .007$). On average, drivers reacted nearly 0.5 s slower to the entering vehicle ($M = 2.4$ s, $SD = 0.4$ s) compared to the other three groups.

When analyzing drivers' velocity chosen to approach the critical intersection situation no significant main effects between the two groups (group with the early-middle warning signal and group



Fig. 5. Mean time and standard deviation (in seconds) while waiting at the intersection with the first critical event, subdivided into the two groups: (1) group with the early-middle warning and (2) group with no warning signal in this epoch (group with the late-middle, the late-sidewise, and no signal, $N = 36$).

without a warning signal in that point in time) were found (60 m: $F_{(1, 46)} = 1.1, p = .292$; 40 m: $F_{(1, 46)} = 0.1, p = .965$; 20 m: $F_{(1, 46)} = 0.7, p = .404$; 10 m: $F_{(1, 46)} = 0.1, p = .785$; 5 m: $F_{(1, 46)} = 2.5, p = .117$).

For the waiting time (Waiting epoch) a significant main effect for the group factor was found ($F_{(1, 46)} = 6.7, p = .013$). Fig. 5 shows the drivers' mean waiting time at the intersection with the first critical event, subdivided into the two groups (group with the early-middle warning signal and group without a warning signal in that point in time).

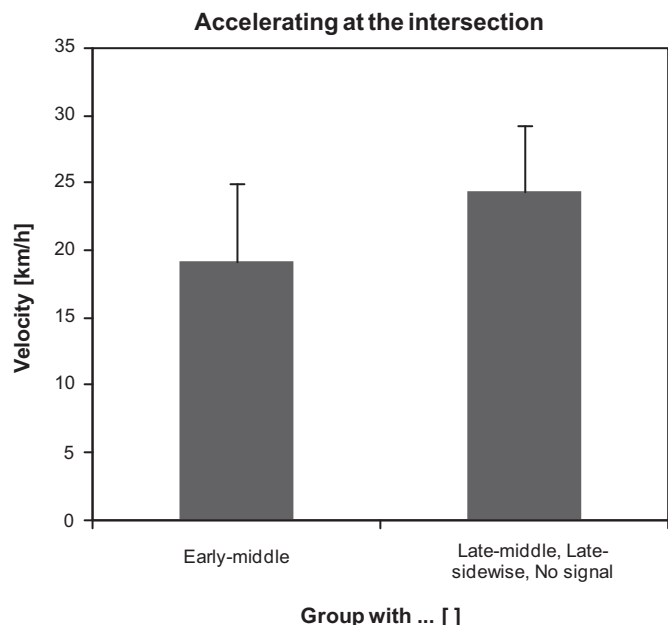


Fig. 6. Mean velocity and standard deviation (in km/h) when turning right at the intersection with the first critical event, subdivided into the two groups: (1) group with the early-middle warning and (2) group with no warning signal at this point (group with the late-middle, the late-sidewise, and no signal, $N = 36$).

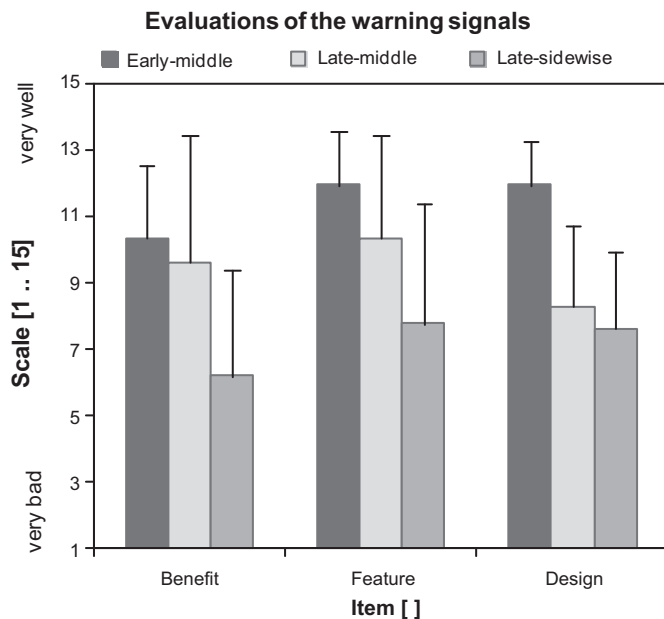


Fig. 7. Subjective evaluations of the three warning signals with regard to their benefit, feature, and design. For the rating, a 15-point rating scale was used (Heller, 1982).

As Fig. 5 shows, drivers waited longer at the intersection when they were already warned about an unexpected hazard (group with the early-middle warning signal), as compared to the other group. Drivers with the early-middle warning signal waited on average 4.7 s (SD = 4.1 s) before the limit line. In the other group the waiting time was approximately 1.8 s (SD = 3.2 s).

For the drivers' mean velocity when entering the intersection (Accelerating point) there was a significant main effect of the group factor ($F_{(1, 46)} = 9.0, p = .004$). In Fig. 6 the results of the mean velocity are shown, subdivided into the two groups (group with the early-middle warning signal and group without a warning signal in that point in time). As Fig. 6 illustrates, the lowest velocity was found in the group with the early-middle warning signal. Drivers turned right with a mean velocity of 19.2 km/h (SD = 5.8 km/h). In contrast, the velocity of the other group (without a warning signal in that point in time) was approximately 24.4 s (SD = 4.9 s).

3.2.2. Evaluation of the warning signals

With regard to the drivers' subjective evaluation of the three warning signals (Fig. 1), the analysis showed a significant main effect of each evaluation criteria (benefit: $F_{(2, 33)} = 5.9, p = .006$; feature: $F_{(2, 33)} = 6.3, p = .005$; design: $F_{(2, 33)} = 15.0, p < .001$). In Fig. 7 the main effects are shown.

As Fig. 7 shows, drivers evaluated the early-middle warning signal as best. They estimated this signal as "useful" ($M = 10.3, SD = 2.2$) and "very good designed" ($M = 11.9, SD = 1.3$). Additionally, the features of this early warning were estimated as "very well" by the drivers ($M = 11.9, SD = 1.6$). Similar patterns were found for the late-middle warning signal (benefit: $M = 9.6, SD = 3.9$; features: $M = 10.3, SD = 3.1$; design: $M = 8.3, SD = 2.5$). In contrast, drivers evaluated the late-sidewise warning signal presented in their peripheral vision as "less useful" ($M = 6.2, SD = 3.2$) and "middle good" with regard to its design ($M = 7.6, SD = 2.3$) and features ($M = 7.8, SD = 3.6$).

4. Discussion

In the study presented here, the effectiveness of three visual warning signals when driving at critical intersections was examined. This was done (1) by giving a warning well in advance in

order to change drivers' expectations about the current driving situation, (2) by presenting a warning in a HUD in drivers' view in order to elicit a fast reaction and (3) by presenting a warning in a HUD at the location of the entering vehicle in order to re-direct drivers' attention toward this unexpected hazard. In the study, a clear positive effect of the first warning signal was found. There was a trend of fewer collisions with this warning signal in the critical intersection situation. All except of one driver were able to avoid a collision with this warning signal. This effect might even have been stronger if the situation had been more critical as only 33% of the drivers of the control group had a collision. However, this early warning was very effective to adapt drivers' behavior toward safer driving. They waited longer and accelerated slower when turning right which enabled them to stop with a much larger minimum of distance toward the entering vehicle. Thus, drivers had and took more time to brake for the critical event in order to avoid a collision. This might also indicate that the critical situation was less strenuous for them. Additionally, drivers evaluated this early warning signal as being most useful in this critical situation.

In contrast, the two later warnings (late-middle and late-sidewise warning signal) were not effective in improving drivers' reaction in the critical event. The number of collisions did not differ from the control group. From the drivers' point of view, these warning signals were presented too late and were not as useful as the early-middle warning signal. It could be that especially the late-middle warning signal distracted the drivers as the warning appeared at a position different from the driver. It may also be that the late-sidewise warning did not provide any additional information to the entering vehicle. Similarly, in a study of Muhrer et al. (2011) a visual warning signal did not prove effective when a car in front was braking suddenly. In this unexpected situation, the brake lights of the front car provided sufficient warning for drivers so that an additional in-vehicle warning system was not beneficial for them.

From a technical point of view, the preferable early warning strategy (top-down processes) requires that information is already be available when drivers are approaching the intersection or are waiting there. This kind of information may be provided by car-to-car technology (e.g. broadcasting the information that a car will enter the road in the near future) or car-to-infrastructure technology (e.g. detecting by a video monitoring of the intersection that a car is likely to enter the road and broadcasting this information). In these cases, early top-down warning signals could be helpful for drivers to re-anticipate the current driving situation, and thus to adapt their driving behavior appropriate to the situation.

The two late warning signals do not seem to be effective as they only provide a warning at the same time when drivers are also able to see the hazard themselves. These warnings could perhaps be more effective if they are given at least to some extent in advance. Even small temporal gains which lead to earlier reactions by drivers might help at least to mitigate a collision. However, the study presented here is not able to give indications about how much advance time would be needed. Given the difficulties described above to provide early warnings, it would be interesting to examine this in more detail by presenting late warnings with very short advance time (e.g. 300 ms before the critical vehicle starts to enter the road). It could also be that drivers would profit more from these warnings if they are given in another modality (e.g. acoustic or haptic) which might lead to faster reactions by drivers (see, e.g. Brown et al., 2005; Lee et al., 2004). Finally, if early top-down warnings are not possible and stronger bottom-up warnings are not efficient, it might also be necessary to introduce autonomous braking of appropriate driver assistance systems. This has been shown to be effective in cases of rear-end accidents (see, e.g. Muhrer et al., 2011) where visual and acoustic warnings were not able to prevent these collisions.

From a methodological point of view, this experimental approach to provoke collisions in order to examine the effectiveness of different warning strategies has proven to be effective. To this aim, a critical intersection situation with a high collision frequency was taken from a previous driving simulator study by Werneke and Vollrath (2012). However, the development of such situations requires some effort and even small variations may change the results. In the previous study (Werneke and Vollrath, 2012), 80% of the drivers (overall $n = 10$ in this specific intersection type) had a collision compared to 33% in this study presented here. The only difference was the color of the entering vehicle which was changed from gray in the previous study (Werneke and Vollrath, 2012) to black in order to achieve a better contrast with the late-sidewise warning signal (Fig. 2). However, the black entering vehicle may have been better visible for drivers against the gray background of the road. Thus, this approach requires a very thorough design of critical situations to be able to examine effects of warning systems.

From a psychological point of view, the results of the study presented here support the idea that driver errors leading to collisions at intersections are at least in some part the results of improper driving behavior according to the driving situation. As shown in the previous study (Werneke and Vollrath, 2012), drivers made more collisions at this intersection type by neglecting task-relevant intersection areas and by turning right with a high velocity. In the study presented here, the analysis of warning strategies showed a positive effect of the early warning signal where drivers waited longer at the intersection before turning right and turned with a lower velocity. The next question which arises here to focus on is what did the drivers do while waiting? One explanation could be that they could have spent more time attempting to locate the reason for the warning by scanning the situation. In order to examine this idea more closely, drivers' gaze behavior should be examined in a next step. Furthermore, it would also be interesting to see whether the early warnings are also effective if they do not include any information of the position of the critical object (e.g. by using a warning triangle instead of a warning circle; see Fig. 2). If the effect of these warnings was to produce a more careful driving behavior, an unspecific warning should be as effective as one including a position information.

Finally, due to the missing motion and the somewhat artificial surrounding in the driving simulator the findings should be validated in real road-traffic where the surrounding is probably more complex and dynamic than in the driving simulator. It would also be interesting to examine this situation and the different warning strategies with older drivers as these driver groups have difficulties in their proper perception of information when driving at intersections (see, e.g. Bao and Boyle, 2007, 2009; Braitman et al., 2007; Caird et al., 2008; Chisholm et al., 2008; McGwin and Brown, 1999; Underwood et al., 2003). In the study presented here, the participants ranged in their age between 21 and 56 years. In the context of age-related differences in intersection negotiation, the age range could be really large. However, accident analyses showed that the highest risk was found under an age of 21 years, and then again over 70 and 75 years (see, e.g. Preusser et al., 1998; Staplin and Lyles, 1991; USDOT, 1996 mentioned in Cobb and Coughlin, 1998; Vollrath, 2007). As perception (see, e.g. Caird and Hancock, 2002; Schiff et al., 1992; Staplin, 1995) and attention errors (see, e.g. Owsley, 2004) are assumed to be the main reasons why older drivers are involved in accidents, warning systems which assist drivers in their attention allocation might be especially useful for these groups of drivers. However, it could also be that warning signals are an additional workload for them which might further contribute to their difficulties at intersections. Thus, experimental studies are clearly necessary for these groups of drivers.

Overall, the study provides a somewhat mixed result. The warning strategy that is most effective is the one which is most difficult

to implement as in most situations it will be very hard to predict some seconds in advance that a critical situation might occur. In contrast, the more realistic warning strategies did not produce a positive effect. However, there might be a chance to improve these strategies by either presenting them at least some milliseconds in advance or by using stronger warning signals in order to elicit faster drivers' behavior. It will be interesting to see future solutions for these situations which are clearly required as these kinds of accidents will probably become even more frequent in the future given the increasing urban traffic.

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